

Nastassia Grimm | Curriculum Vitae

Royal Society Newton International Fellow

Department of Physics, University of Oxford

Denys Wilkinson Building, Oxford OX1 3RH, United Kingdom

✉ nastassia.grimm@physics.ox.ac.uk

💻 Personal website: <https://nastassiag.github.io>

Key interests and achievements: My research is focused on testing the laws of gravity at cosmological scales and on general relativistic effects. I am currently preparing for upcoming cosmological data releases by stage-IV surveys such as *Euclid* as an active member of the *Euclid Consortium*. Moreover, I am interested in pulsar timing arrays and gravitational waves, specifically in their synergies with other probes of our Universe. I have a strong background in mathematics as well as being highly experienced in coding and data analysis. I currently hold a [Newton International Fellowship](#) by the *Royal Society*, a fellowship with a 3% success rate designed to support outstanding early career researchers. Further highlights include a bronze medal from the [International Mathematical Olympiad](#), as well as recent first-authored publications in [Nature Communications](#) and [Physical Review Letters](#). Moreover, I am highly dedicated to teaching and outreach.

Personal Information

Date of birth 12 May 1994

Citizenship Germany

Employment

31/03/2026 to present* **Royal Society Newton International Fellow** at *Department of Physics, University of Oxford*, Oxford, United Kingdom.

*Fellowship funded until 30/03/2028

18/08/2025 to 30/03/2026 **Research Fellow** at *Institute of Cosmology and Gravitation, University of Portsmouth*, Portsmouth, United Kingdom.

01/10/2021 to 31/07/2025* **Postdoctoral researcher** at *Département de Physique Théorique, University of Geneva*, Geneva, Switzerland.

*Excluding the period 05/03/2023 to 03/09/2023 (maternity leave).

Education

01/09/2017 to 17/09/2021 **PhD student**, *Center for Theoretical Astrophysics and Cosmology, University of Zurich*, Zurich, Switzerland.

Supervisor: Prof. Dr Jaiyul Yoo

Thesis title: "General relativistic effects in cosmological weak lensing"

14/09/2015 **MSc**, *ETH Zurich*, Zurich, Switzerland.

to 28/08/2017 Master of Science in Applied Mathematics; Area of application: Theoretical Physics
Graduated with total grade 5.57 (best possible: 6.0; lowest passing grade: 4.0)

17/09/2012 **BSc**, *University of Zurich*, Zurich, Switzerland.

to 21/09/2015 Bachelor of Science in Mathematics, with Physics as a minor subject

Graduated with total grade 5.5 (best possible: 6.0; lowest passing grade: 4.0)

Total grade major subject: 5.4; Total grade minor subject: 5.8

Awarded research funding

Award date: **Royal Society Newton International Fellowships**

19/09/2025 Project name: *Galaxy surveys in the present and future: Testing gravity with Euclid and exploring synergies with Pulsar Timing Arrays*

Awarded funding: 290'472 GBP

Experience in teaching & mentoring

- Nov. 2025 Teaching at PhD-level, *University of Portsmouth*, Portsmouth, United Kingdom.
Responsibilities: Design and delivery of a one-week intensive course on “Theoretical Cosmology” for first-year PhD students.
- Oct. 2021 Teaching & mentoring activities, *University of Geneva*, Geneva, Switzerland.
to July 2025 Responsibilities: Teaching assistance for Bachelor and Master courses, mentoring of PhD students, Master thesis referee and co-examiner.
- Fall 2017 Teaching assistance, *University of Zurich*, Zurich, Switzerland.
to fall 2020 Courses: “The Universe”, “Introduction to Astrobiology”, “Theoretical Astrophysics”, “Theoretical Cosmology” and “Introduction to Astrophysics”.
Responsibilities: Preparation of exercise sheets and solutions, correction of exercises, leading practice sessions, serving as co-examiner.
- Fall 2015 Teaching assistance, *ETH Zurich*, Zurich, Switzerland.
to fall 2016 Courses: “Analysis II” and “Analysis III”.
Responsibilities: Correction of exercise sheets, leading practice sessions.
- Fall 2013 Teaching assistance, *University of Zurich*, Zurich, Switzerland.
Course: “Programming for Mathematicians” (programming language: Java).
Responsibilities: Correction of exercise sheets, leading practice sessions.

Selection of conference, workshop and seminar talks

- 25/06/2026 Invited talk on “Model-independent tests of gravity with the Weyl potential evolution” at the *31st International Symposium on Particles, Strings, and Cosmology* (Sheffield, United Kingdom)
- 26/03/2025 Plenary talk on “A model-independent test of gravity from the Weyl potential evolution” at the *Euclid Consortium meeting 2025* (Leiden, Netherlands).
- 11/07/2024 Presentation on “Combining the Weyl potential and galaxy velocities: new measurements of E_G ” at the workshop *Relativistic Effects and Novel Observables in Cosmology* (University of Geneva).
- 14/05/2024 Presentation on “A model-independent test of gravity from the Weyl potential evolution” at the *4th EuCAPT Annual Symposium* (CERN).
- 21/06/2022 Presentation on “Rescuing constraints on modified gravity through relativistic distortions in large-scale structure” at the workshop *General Relativistic effects in observing the Large-Scale Structure of the Universe* (Centro de Astrofísica da Universidade do Porto).
- 03/05/2021 Presentation on “Galaxy Power Spectrum in General Relativity” at the workshop *Relativistic Aspects of Large-Scale Structure - Theory and Simulation* (University of Zurich). The talk can be found online: <https://youtu.be/0pEXngDZigg>
- June to Nov. 2020 Series of invited seminar talks on “Galaxy Power Spectrum in General Relativity” at
- CERN, Theory Department (11/06/2020).
- California Institute of Technology (06/10/2020).
- University of Cambridge, Institute of Astronomy (12/10/2020).
- University of Oxford, Department of Physics (10/11/2020).
- Princeton University / Institute for Advanced Study (30/11/2020).
- 06/02/2018 Presentation on “Jacobi Mapping Approach for a precise Cosmological Weak Lensing Formalism” at the *Swiss Cosmology Days 2018* (CERN).

Outreach and Public Engagement

- 30/06/2026 Team member of the *Euclid, the dark universe detective* flagship exhibit at the Royal Society’s [Summer Science Exhibition 2026](#), reaching a total of 10’815 visitors.
- 28/01/2026 Outreach talk at [Stargazing 2026](#) at the *University of Portsmouth*.

- 30/10/2025 Outreach talk at the [Dark Matter Day](#) at the *University of Oxford*.
- 2022 to 2024 Contributions to the YouTube channel [Cosmic Blueshift](#) in French & English, notably:
- [Will Einstein pass the test of space-time distortions?](#) [Original title: *Einstein passera-t'il le test des distorsions de l'espace-temps?*];
Published 03/04/2024; 6,608 views as of 07/07/2026.
 - [Measuring spacetime distortions with light bending](#) [Original title: *Mesurer les déformations de l'espace-temps avec la déviation de la lumière*];
Published 28/12/2023; 18,675 views as of 07/07/2026.
 - [Modifying Einstein's theory of gravity](#) [Original title: *Modifier la théorie de la gravité d'Einstein*];
Published 01/12/2023, 11,073 views as of 07/07/2026.
- 07/02/2024 Outreach talk at the *Institut International de Lancy* as part of the International Day of Women and Girls in Science (including an [interview in the school web radio](#)); student age: 14-15 years.
- 02/02/2023 Outreach talk at the *Ecole Internationale de Genève* as part of the International Day of Women and Girls in Science; student age: 12-13 years.

Organisation of seminars and conferences

- January 2027 (planned) Member of the organising committee of the *CosmoFONDUE 2027 (Cosmological Fundamental Observables and Novel Discoveries in Universe Evolution)* conference at *CERN*; the organisation of this event has been granted 10,000 CHF by the *CERN theory department*.
- Aug. 2025 to March 2026 Main organiser of the *Theoretical Cosmology Seminars* at the *Institute of Cosmology and Gravitation, University of Portsmouth*.
- June 2025 Member of the organising committee of the *CosmoFONDUE 2026* conference at the *University of Geneva*; the *Société académique de Genève* has funded this event with 10,000 CHF and the *Swiss National Science foundation* with an additional 2,500 CHF, following our successful proposal to which I made major contributions.
- Sept. 2023 to July 2025 Main organiser of the *Cosmology and Particle Physics Seminar Series* at the *University of Geneva*.

Awards and achievements at mathematical competitions

- July 2012 Awarded a bronze medal at the 53rd International Mathematical Olympiad in Mar del Plata (Argentina).
- May 2012 Achieved the 5th place at the Austrian national competition of the Mathematical Olympiad.
- May 2011 Achieved the 11th place at the Austrian national competition of the Mathematical Olympiad; Qualified for the 5th Middle European Mathematical Olympiad in Varaždin (Croatia) and represented Austria.
- 2007 Achieved the 3rd place of all Austrian participants in the "Mathematical Kangaroo" (age grade 7).

Software products

- 2026 Contributor to `cloelib` and `cloelike`.
Publicly available at: <https://github.com/cloe-org>
Description: `cloelib` is a flexible and efficient library designed to compute cosmological observables for the CLOE (Cosmology Likelihood for Observables in Euclid) project. `cloelike` is the module for Euclid primary observable theoretical predictions, interfacing with `cloelib`.
Scientific usage: `cloelib` and `cloelike` provide the foundation for cosmological data analysis of upcoming Euclid Data Releases, with tools and methods being designed to be broadly applicable across current and future cosmological experiments.

- 2023 Co-development of the Python code EF-TIGRE (*Effective Field Theory of Interacting dark energy with Gravitational REdshift*).
Publicly available at: <https://github.com/Mik3M4n/EF-TIGRE>
Description: Package for constraining interacting dark energy/dark matter models in the effective field theory framework through large-scale structure observables, including the gravitational redshift effect.
Scientific usage: Used in Castello et al. (2024), [arXiv:2311.14425](https://arxiv.org/abs/2311.14425), and Tutusaus et al. (2024), [arXiv:2312.06434](https://arxiv.org/abs/2312.06434).

IT Skills

- Python Advanced; frequent application and development of software for own research work.
Mathematica Advanced; applied for numerical evaluations in own research work.
MATLAB Trained in machine learning applications (completed online course by Stanford University).
Java Completed a respective university course in 2012 with grade 6.0; Teaching assistant for the same course in the year 2013.

Languages

- German Native speaker
English Fluent (level C2, certified by the “Cambridge Certificate in Advanced English”)
French, Russian Intermediate